Modeling How Macroeconomic Shocks Affect Regional Employment: Analyzing the Brazilian Formal Labor Market using the Global VAR Approach

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Talking points

Talking Points

Motivation

- The assessment of economic variables such as unemployment
  - Okun's Law: Unemployment rate is directly associated with GDP;
- Market Integration
  - Financial and economic interdependence between regions;
  - Macroeconomic analyzes should consider interdependencies.;
- Job Market
  - Macroeconomic level: influenced by the dynamics of the domestic market and by the dynamics of the external market.;
  - Microeconomic level: interaction between companies and individual employees.

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- Focus on the Brazilian labor market
  - Establish a model for the Brazilian labor market, at the regional level, which takes into account the interdependencies between the Brazilian regions.
  - Estimate the regional elasticity of employment in relation to the country's economic activity;

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Brief description of articles on Brazilian Formal Labor Market

- Oliveira and Carneiro (2001) - Analysis of the fluctuations in the employment level of several Brazilian states in relation to national employment with the objective of verifying a long term relationship.
- Camila Kraide Kretzmann and Marina Silva da Cunha (2009) - Analysis of the formal labor market between metropolitan and non-metropolitan areas in Brazil.
- Norbert Schanne (2015) - Analysis of the regional labor market in Germany, focusing on forecasting regional labor markets. It used the GVAR methodology and used it to perform a prospective analysis of the German economy.From many intertemporal macroeconomic models it's possible to obtain the following steady state equations

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A brief description of GVAR approach

- The introduction of the global autoregressive vector (GVAR) methodology was made by Pesaran, Weiner and Schuermann (2004)[1], and improved on by Dees, Mauro, Pesaran and Smith (2007)
- Chudik and Pesaran (2011) - Analyzed the curse of dimensionality in the case of large dynamic linear systems. The infinite-dimensional VAR could be well approximated by a set of finite-dimensional small-scale models that can be consistently estimated separately.

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Talking Points

The Global VAR model as a two-step procedure

- First stage: Specific models of each region are estimated conditioned to the external influences of the other regions.

$$x_i t = a_{i0} + a_{i1} t + \sum_{j=1}^{p_i} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x_{it}^* + \sum_{l=1}^{qi} \Lambda_{il} x_{i,t-l} + u_{it}$$

$$x_{it}^* = \sum_{j=0}^{N} W_{ij} x_{jt}$$

$$A_{i0} W_i x_t = a_{i0} + a_{i1} t + \sum_{l=1}^{p} A_{il} W_i x_{t-l} + \varepsilon_t$$

The Global VAR model as a two-step procedure

- Second Stage: The VAR models of each region are grouped into a single Global VAR model.

$$G_0 = \begin{bmatrix} A_{00}W_0 \\ A_{10}W_1 \\ \vdots \\ A_{N0}W_N \end{bmatrix}; \quad G_l = \begin{bmatrix} A_{0l}W_0 \\ A_{1l}W_1 \\ \vdots \\ A_{Nl}W_N \end{bmatrix}; a_0 = \begin{bmatrix} a_{00} \\ a_{10} \\ \vdots \\ a_{N0} \end{bmatrix}; \quad a_1 = \begin{bmatrix} a_{01} \\ a_{11} \\ \vdots \\ a_{N1} \end{bmatrix}; \quad \varepsilon_1 = \begin{bmatrix} \varepsilon_{0t} \\ \varepsilon_{1t} \\ \vdots \\ \varepsilon_{Nt} \end{bmatrix};$$

$$G_0 x_t = a_0 + a_1 t + \sum_{l=1}^{p} G_l x_{t-l} + \varepsilon_t$$

$$F_i = G_0^{-1} G_i; \quad b_0 = G_0^{-1} a_0; \quad b_1 = G_0^{-1} a_1; \quad e_t = G_0^{-1} \varepsilon_t;$$

$$x_t = b_o + b_1 t + \sum F_i x_{t-i} + e_t$$

Model

- The regions used in the model will be the mesoregions defined by the Brazilian Institute of Geography and Statistics in 2015.
- The labor market model will consist of workers admitted and disconnected from the labor market.

$$x_{it} = \begin{bmatrix} Adm_{it} \\ Dis_{it} \end{bmatrix}; \quad x_{it}^* = \begin{bmatrix} Adm_{it}^* \\ Dis_{it}^* \end{bmatrix}$$

$$Adm_{it}^* = \sum_{j=1}^{N} W_{ij} Adm_{jt} \quad Dis_{it}^* = \sum_{j=1}^{N} W_{ij} Dis_{jt}$$

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Model

- Macroeconomic variables are modeled in a Dominant Unit.
- The Dominant Unit will contain the logarithmic level of Brazilian industrial production with feedback from the regions:

$$\omega_t = [ln(IndProd)]$$

$$\Delta\omega_t = \Phi_{\omega 0} + \sum_{l=1}^{k1} \Phi_{\omega l}\Delta\omega_{t-l} + \sum_{l=1}^{k_2} \Lambda_{\omega l}\Delta x^*_{i,t-l} + \eta_{\omega t}$$

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Weight matrix

- The weight matrix should capture the importance of all other regions for the i region (DI MAURO, PESARAN, 2013).
- Determining the importance of one region to another is to determine the importance of one labor market to another labor market.
    - Each region has its own labor market.
        - Workers freely exchange markets.
        - For a worker to move to another region there must be a connection between those regions.
    - The importance of region j for region i will be a function of the connections between them.

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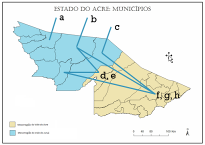


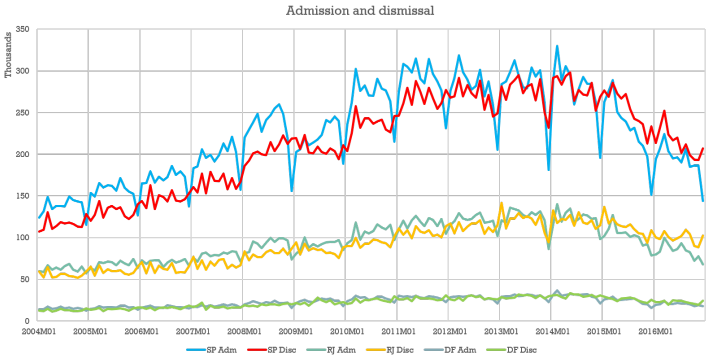
$$W_{blue} = \frac{\begin{bmatrix} c_{i1} & \dots & c_{iN} \end{bmatrix}}{\sum_{j=1}^{N} c_{ij}} \qquad W_{blue} = \frac{\begin{bmatrix} 5 & 3 \end{bmatrix}}{8} = \begin{bmatrix} 0.625 & 0.375 \end{bmatrix}$$

Talking Points

Data

Variable	Source	Frequency	Range
Brazilian Industrial Production	IBGE	Monthly	2004-01 to 2016-12
Employment and Unemployment	CAGED	Monthly	2004-01 to 2016-12
Meso-regions and municipalities	IBGE	N/A	as of 2015
Connecting between regions	IBGE	N/A	as of 2007

Data


Admission and dismissal
Thousands
350
300
250
200
150
100
50
0
2004M01
2005M01
2006M01
2007M01
2008M01
2009M01
2010M01
2011M01
2012M01
2013M01
2014M01
2015M01
2016M01
SP Adm
SP Disc
RJ Adm
RJ Disc
DF Adm
DF Disc

Talking Points

Results

- 137 specific models were specified.
  - Domestic variables: 98 models with lag 2 and 39 models with lag 1
  - Foreign variables: 91 models with lag 2 and 46 models with lag 1.
  - 9 models with 1 cointegration relation and 128 models with 2 cointegration relations.
- Several scenarios were estimated to assess the impact of industrial production on the labor market.
  - Annual growth of 0%: A negative employment balance for at least the next 8 years.
  - Annual growth of 2%: A positive accumulated balance of 2.5 million job positions for 8 years ahead.
  - Higher growth scenarios (4%, 6% and 8%) show significant improvements in the labor market.

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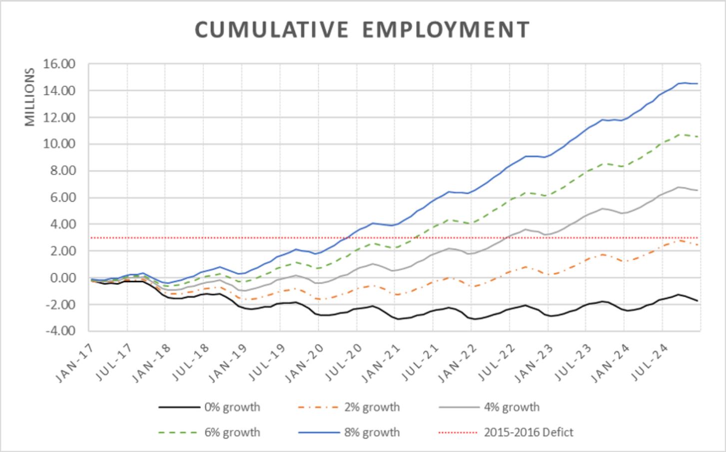
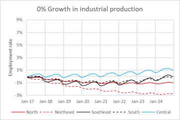
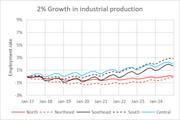
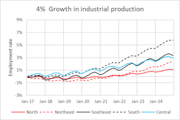

CUMULATIVE EMPLOYMENT
MILLIONS
16.00
14.00
12.00
10.00
8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
JAN-17
JUL-17
JAN-18
JUL-18
JAN-19
JUL-19
JAN-20
JUL-20
JAN-21
JUL-21
JAN-22
JUL-22
JAN-23
JUL-23
JAN-24
JUL-24
0% growth
2% growth
4% growth
6% growth
8% growth
2015-2016 Defict

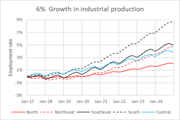
5 major regions

- Focusing on the five major regions of Brazil (North, Northeast, Southeast, South and Central), it was possible to determine the elasticity of employment in relation to industrial production for each major region. South (0.18) - North East (0.16) - South East (0.1) - North (0.06) - Central (0.06)



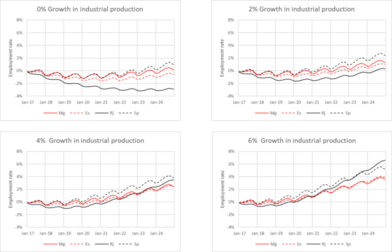






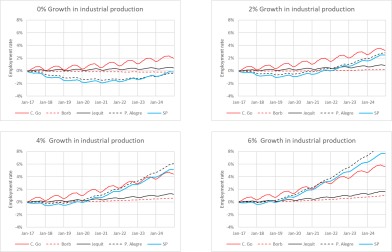
Southeast states

- Analyzing the states of the Southeast we have: Rio de janeiro (0.19) - Espirito Santo (0.09) - São Paulo (0.08) - Minas Gerais (0.07)



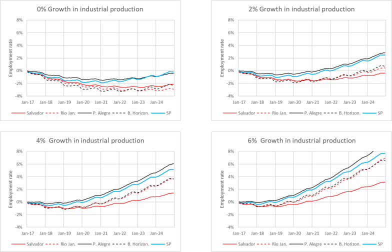
Meso-regions

- The GVAR methodology allows estimation of the elasticity of each meso-region. Porto Alegre (0.19) - São Paulo (0.16) - C. Goiás (0.07) - Borborema (0.02) - Jequit. (0.02)



Meso-regions

- Elasticity of the most populated meso-regions. Rio de Janeiro (0.19) - Porto Alegre (0.19) - Belo Horizonte (0.17) - São Paulo (0.16) - C. Goiás (0.07) - Salvador (0.11)



Dynamics of the region

- VIDEO

Talking Points

Main Conclusions and limitations

- The GVAR methodology proved a parsimonious modeling of the labor market at regional level, (mesoregion) taking into account the external influences of each region.
- The model showed the dynamic relationship between the mesoregions, as well as it was possible to determine the elasticity of the regions.
- The South, Southeast and Northeast have shown to be very elastic, presenting a constant interaction in their regions, while the North and Central regions seem to suffer little influence from the neighboring regions.
- Limitations
  - Small sample of employment series, data availiable only from 2004 onwards.
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Assets and Liabilities returns:

- A more comprehensive macroeconomic model can be formulated and different types of macroeconomic shocks can be identified and their impact simulated.
- It is possible to construct a "tailor-made" GVAR. This route was explored by Ericsson and Reisman [2012] but not in a regional model.
- Disaggregate data by not only regions but also sectors such as agriculture, industry and services among others.
- Another possible path of research consists of evaluating predictive performance at different aggregate levels and horizons against simple benchmarks such as random walk or first order autoregressive model.

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M Hashem Pesaran, Til Schuermann, and Scott M Weiner.
Modeling regional interdependencies using a global error-correcting macroeconometric model.
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